LINMA2710 - Scientific Computing Single Instruction Multiple Data (SIMD)

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☐ Full Width Mode ☐ Present Mode

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Motivation

The need for parallelism

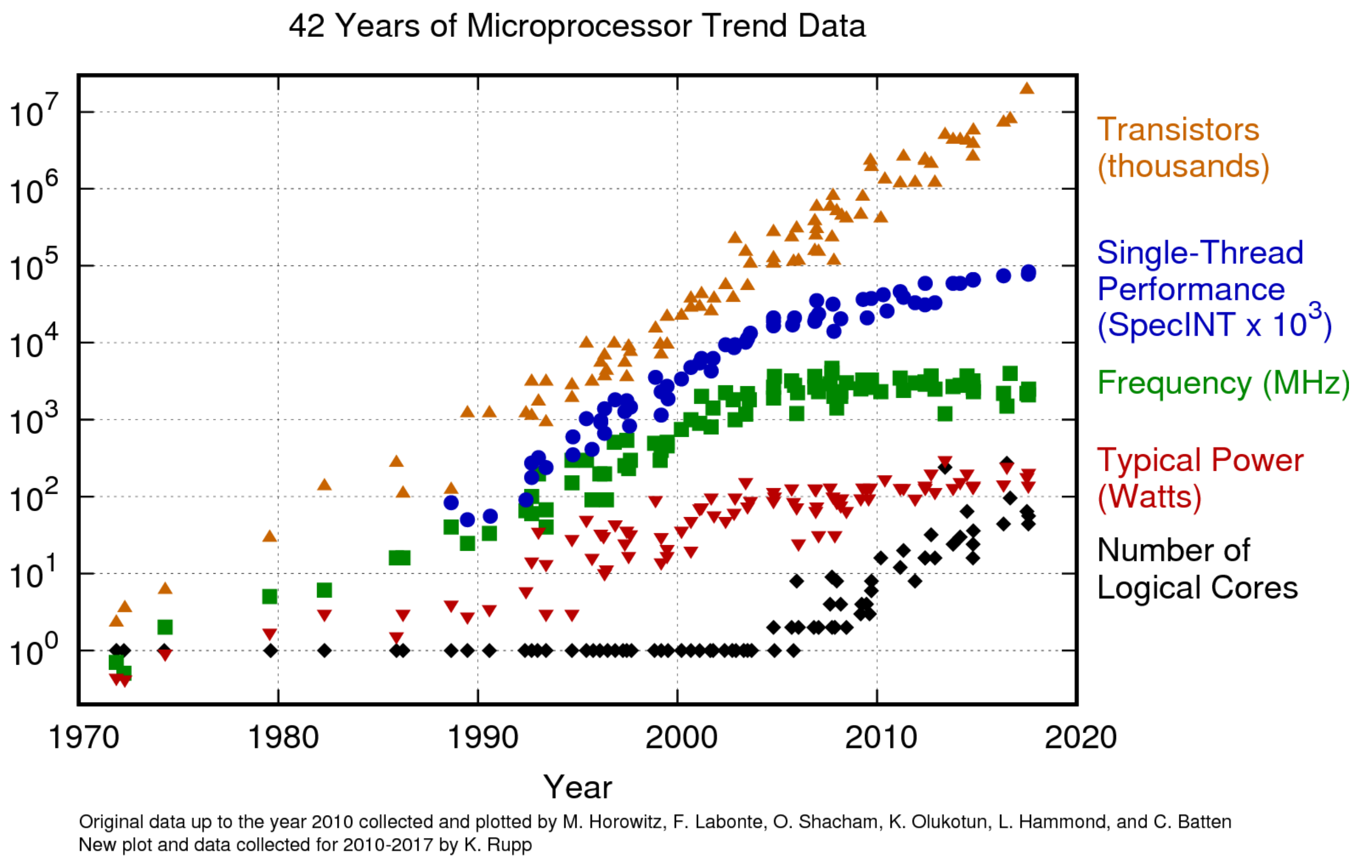
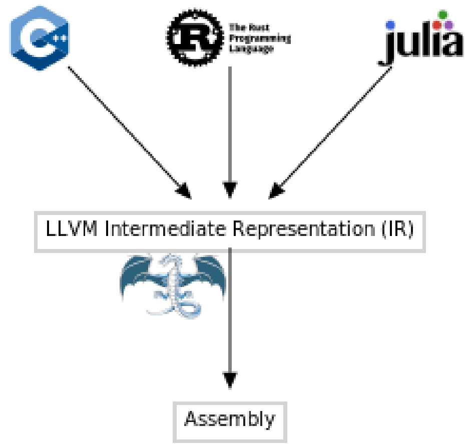


Image source

A bit of historical context

- **1972** : C language created by Dennis Ritchie and Ken Thompson to ease development of Unix (previously developed in **assembly**)
- **1985** : C++ created by Bjarne Stroustrup
- **2003** : LLVM started at University of Illinois
- **2005** : Apple hires Chris Lattner from the university
- **2007** : He then creates the LLVM-based compiler Clang
- **2009** : Mozilla start developing an LLVM-based compiler for Rust
- **2009** : Development starts on Julia, with LLVM-based compiler



A sum function in C and Julia

```c
float sum(float *vec, int length) {
    float total = 0;
    for (int i = 0; i < length; i++) {
        total += vec[i];
    }
    return total;
}
```

```julia
c_sum(x::Vector{Cfloat}) = ccall(("sum", sum_float_lib), Cfloat, (Ptr{Cfloat}, Cint), x, length(x));
```

```
julia_sum (generic function with 1 method)
```

```julia
function julia_sum(v::Vector{T}) where {T}
    total = zero(T)
    for i in eachindex(v)
        total += v[i]
    end
    return total
end
```

Let's make a small benchmark

```
vec_float =
▸[0.152564, 0.943411, 0.806974, 0.617942, 0.770227, 0.454559, 0.832874, 0.194885, 0.668397
```

```julia
vec_float = rand(Float32, 2^16)
```

```
32775.977f0
```

```julia
@btime c_sum($vec_float)
```

```
242.514 μs (0 allocations: 0 bytes)
```

```
32775.977f0
```

```julia
@btime julia_sum($vec_float)
```

```
60.633 μs (0 allocations: 0 bytes)
```

▸ **How to speed up the C code ?**

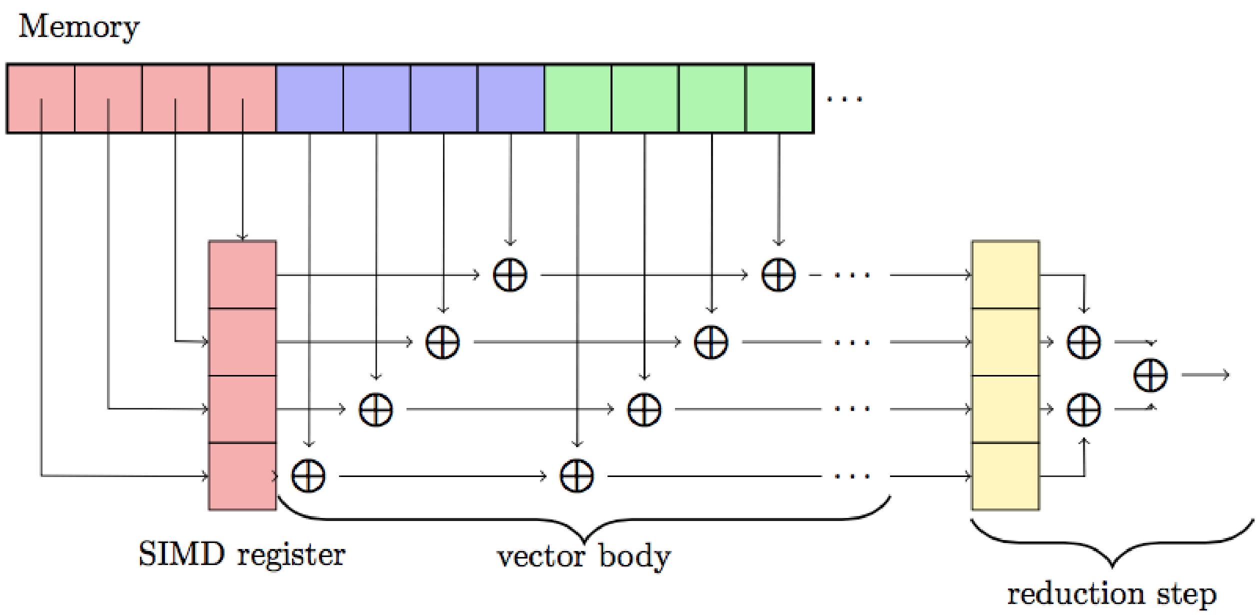
```
⚠ PlutoRunner.embed_display(x) is deprecated and will be removed in a future rele
ase. Use AbstractPlutoDingetjes.Display.@embed(x) instead.
```

Tip

As accessing global variables is slow in Julia, it is important to add `$` in front of them when using `btime`. This is less critical in Pluto though as it handles global variables differently. To see why, try removing the `$`, you should see `1` allocations instead of zero.

```c
float sum(float *vec, int length) {
    float total = 0;
    for (int i = 0; i < length; i++) {
        total += vec[i];
    }
    return total;
}
```

Summing with SIMD



Faster Julia code

▶ **How to get the same speed up from the Julia code ?**

⚠ PlutoRunner.embed_display(x) is deprecated and will be removed in a future release. Use AbstractPlutoDingetjes.Display.@embed(x) instead.

```
julia_sum_fast (generic function with 1 method)
```

```julia
function julia_sum_fast(v::Vector{T}) where {T}
    total = zero(T)
    for i in eachindex(v)
        @fastmath total += @inbounds v[i]
    end
    return total
end
```

32775.996f0

```
@btime julia_sum_fast($vec_float)
```

```
2.544 μs (0 allocations: 0 bytes)
```

julia_sum_simd (generic function with 1 method)

```
function julia_sum_simd(v::Vector{T}) where {T}
    total = zero(T)
    @simd for i in eachindex(v)
        total += v[i]
    end
    return total
end
```

32775.996f0

```
@btime julia_sum_simd($vec_float)
```

```
2.544 μs (0 allocations: 0 bytes)
```

Careful with fast math

▶ **Why are the three elements in the center of the vector ignored in this example ?**

```
⚠ PlutoRunner.embed_display(x) is deprecated and will be removed in a future rele
ase. Use AbstractPlutoDingetjes.Display.@embed(x) instead.
```

test_kahan = ▶[1.0, 2.98023f-8, 2.98023f-8, 2.98023f-8, 0.000119209]

```
test_kahan = Cfloat[1.0, eps(Cfloat)/4, eps(Cfloat)/4, eps(Cfloat)/4,
1000eps(Cfloat)]
```

1.000119298696518

```
sum(Float64.(test_kahan))
```

1.0001192f0

```
c_sum(test_kahan[[1, 5]])
```

1.0001192f0

```
c_sum(test_kahan)
```

To improve the accuracy this, we consider the Kahan summation algorithm.

```
1.0001193f0
c_sum_kahan(test_kahan)
```

Optimization level : -O0 Enable `-ffast-math` ?

▶ **What happens when `-ffast-math` is enabled ?**

```
⚠ PlutoRunner.embed_display(x) is deprecated and will be removed in a future rele
ase. Use AbstractPlutoDingetjes.Display.@embed(x) instead.
```

For further details, see this blog post.

Tip

`eps` gives the difference between `1` and the number closest to `1`. See also `prevfloat` and `nextfloat`.

```c
float sum_kahan(float* vec, int length) {
    float total, c, t, y;
    int i;
    total = c = 0.0f;
    for (i = 0; i < length; i++) {
      y = vec[i] - c;
      t = total + y;
      c = (t - total) - y;
      total = t;
   }
   return total;
}
```

SIMD inspection

Instruction sets

The data is **packed** on a single SIMD unit whose width and register depends on the instruction set family. The single instruction is then run in parallel on all elements of this small **vector** stored in the SIMD unit. These give the prefix `vp` to the instruction names that stands from *Vectorized Packed*.

Instruction Set Family	Width of SIMD unit	Register
Streaming SIMD Extension (SSE)	128-bit	%xmm
Advanced Vector Extensions (AVX)	256-bit	%ymm
AVX-512	512-bit	%zmm

```
ProcessChain([Process(`lscpu`, ProcessExited(0)), Process(`grep Flag`, ProcessExited(0))
```

```
run(pipeline(`lscpu`, `grep Flag`))
```

```
Flags:                            fpu vme de pse tsc msr pae mce cx8
apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx m
mxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable nonst
op_tsc cpuid extd_apicid aperfmperf tsc_known_freq pni pclmulqdq ssse3 fma cx1
6 pcid sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm
cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topoext vmm
call fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves user_shstk clzero xsaveerptr rdpru arat
npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthr
eshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm
```

Tip

To determine which instruction set is supported for your computer, look at the `Flags` list in the output of `lscpu`. We can check in the Intel® Intrinsics Guide that `avx`, `avx2` and `avx_vnni` are in the AVX family.

SIMD at LLVM level

How can you check that SIMD is enabled? Let's check at the level of LLVM IR.

```
f (generic function with 1 method)
```

```julia
function f(x1, x2, x3, x4, y1, y2, y3, y4)
    z1 = x1 + y1
    z2 = x2 + y2
    z3 = x3 + y3
    z4 = x4 + y4
    return z1, z2, z3, z4
end
```

```julia
@code_llvm debuginfo=:none f(1, 2, 3, 4, 5, 6, 7, 8)
```

```
; Function Signature: f(Int64, Int64, Int64, Int64, Int64, Int64, Int64, In
t64)
define void @julia_f_34855(ptr noalias nocapture noundef nonnull sret([4 x i6
4]) align 8 dereferenceable(32) %sret_return, i64 signext %"x1::Int64", i64 si
gnext %"x2::Int64", i64 signext %"x3::Int64", i64 signext %"x4::Int64", i64 si
gnext %"y1::Int64", i64 signext %"y2::Int64", i64 signext %"y3::Int64", i64 si
gnext %"y4::Int64") #0 {
top:
  %0 = add i64 %"y1::Int64", %"x1::Int64"
  %1 = add i64 %"y2::Int64", %"x2::Int64"
  %2 = add i64 %"y3::Int64", %"x3::Int64"
  %3 = add i64 %"y4::Int64", %"x4::Int64"
  store i64 %0, ptr %sret_return, align 8
  %"new::Tuple.sroa.2.0.sret_return.sroa_idx" = getelementptr inbounds i8, ptr
%sret_return, i64 8
  store i64 %1, ptr %"new::Tuple.sroa.2.0.sret_return.sroa_idx", align 8
  %"new::Tuple.sroa.3.0.sret_return.sroa_idx" = getelementptr inbounds i8, ptr
%sret_return, i64 16
  store i64 %2, ptr %"new::Tuple.sroa.3.0.sret_return.sroa_idx", align 8
  %"new::Tuple.sroa.4.0.sret_return.sroa_idx" = getelementptr inbounds i8, ptr
%sret_return, i64 24
  store i64 %3, ptr %"new::Tuple.sroa.4.0.sret_return.sroa_idx", align 8
  ret void
}
```

Tip

If we see `add i64`, it means that each `Int64` is added independently

Packing the data to enable SIMD

```
f_broadcast (generic function with 1 method)
```

```julia
function f_broadcast(x, y)
    z = x .+ y
    return z
end
```

```julia
@code_llvm debuginfo=:none f_broadcast((1, 2, 3, 4), (1, 2, 3, 4))
```

```
; Function Signature: f_broadcast(NTuple{4, Int64}, NTuple{4, Int64})
define void @julia_f_broadcast_33503(ptr noalias nocapture noundef nonnull sret([4 x i64]) align 8 dereferenceable(32) %sret_return, ptr nocapture noundef nonnull readonly align 8 dereferenceable(32) %"x::Tuple", ptr nocapture noundef nonnull readonly align 8 dereferenceable(32) %"y::Tuple") #0 {
top:
  %0 = load <4 x i64>, ptr %"x::Tuple", align 8
  %1 = load <4 x i64>, ptr %"y::Tuple", align 8
  %2 = add <4 x i64> %1, %0
  store <4 x i64> %2, ptr %sret_return, align 8
  ret void
}
```

Tip

`load <4 x i64>` means that 4 `Int64` are loaded into a 256-bit wide SIMD unit.

SIMD at assembly level

```
@code_native debuginfo=:none f_broadcast((1, 2, 3, 4), (1, 2, 3, 4))
```

```
	.text
	.file	"f_broadcast"
	.section	.ltext,"axl",@progbits
	.globl	julia_f_broadcast_33807          # -- Begin function julia_f_broadc
ast_33807
	.p2align	4, 0x90
	.type	julia_f_broadcast_33807,@function
julia_f_broadcast_33807:                # @julia_f_broadcast_33807
; Function Signature: f_broadcast(NTuple{4, Int64}, NTuple{4, Int64})
# %bb.0:                                # %top
	#DEBUG_VALUE: f_broadcast:x <- [$rsi+0]
	#DEBUG_VALUE: f_broadcast:y <- [$rdx+0]
	push	rbp
	vmovdqu ymm0, ymmword ptr [rdx]
	mov rbp, rsp
	mov rax, rdi
	vpaddq	ymm0, ymm0, ymmword ptr [rsi]
	vmovdqu ymmword ptr [rdi], ymm0
	pop rbp
	vzeroupper
	ret
.Lfunc_end0:
	.size	julia_f_broadcast_33807, .Lfunc_end0-julia_f_broadcast_33807
                                        # -- End function
	.type	".L+Core.Tuple#33809",@object   # @"+Core.Tuple#33809"
	.section	.lrodata,"al",@progbits
	.p2align	3, 0x0
".L+Core.Tuple#33809":
```

Tip

The suffix `v` in front of the instruction stands for `vectorized`. It means it is using a SIMD unit.

Tuples implementing the array interface

N = 2

```julia
let
    T = Float64
    A = rand(SMatrix{N,N,T})
    x = rand(SVector{N,T})
    @code_llvm debuginfo=:none A * x
end
```

```
; Function Signature: *(StaticArraysCore.SArray{Tuple{2, 2}, Float64, 2,
4}, StaticArraysCore.SArray{Tuple{2}, Float64, 1, 2})
define void @"julia_*_34800"(ptr noalias nocapture noundef nonnull sret([1 x
[2 x double]]) align 8 dereferenceable(16) %sret_return, ptr nocapture noundef
nonnull readonly align 8 dereferenceable(32) %"A::SArray", ptr nocapture nound
ef nonnull readonly align 8 dereferenceable(16) %"B::SArray") #0 {
top:
  %"A::SArray[3]_ptr" = getelementptr inbounds i8, ptr %"A::SArray", i64 16
  %0 = load <2 x double>, ptr %"B::SArray", align 8
  %1 = load <2 x double>, ptr %"A::SArray", align 8
  %2 = shufflevector <2 x double> %0, <2 x double> poison, <2 x i32> zeroiniti
alizer
  %3 = fmul <2 x double> %1, %2
  %4 = load <2 x double>, ptr %"A::SArray[3]_ptr", align 8
  %5 = shufflevector <2 x double> %0, <2 x double> poison, <2 x i32> <i32 1, i
32 1>
  %6 = fmul contract <2 x double> %4, %5
  %7 = fadd contract <2 x double> %3, %6
  store <2 x double> %7, ptr %sret_return, align 8
  ret void
}
```

Tip

Small arrays that are allocated on the stack like tuples and implemented in `StaticArrays.jl`. Operating on them leverages SIMD.

Auto-Vectorization

LLVM Loop Vectorizer for a C array

```
emit_llvm(c_sum_code_for_llvm, cflags = [sum_opt; sum_flags]);
```

```
; ModuleID = '/tmp/jl_ZOCdT9/main.c'
source_filename = "/tmp/jl_ZOCdT9/main.c"
target datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-i128:128-f80:128-n8:16:32:64-S128"
target triple = "x86_64-unknown-linux-gnu"

; Function Attrs: noinline nounwind optnone uwtable
define dso_local i32 @sum(ptr noundef %0, i32 noundef %1) #0 {
  %3 = alloca ptr, align 8
  %4 = alloca i32, align 4
  %5 = alloca i32, align 4
  %6 = alloca i32, align 4
  store ptr %0, ptr %3, align 8
  store i32 %1, ptr %4, align 4
  store i32 0, ptr %5, align 4
  store i32 0, ptr %6, align 4
  br label %7

7:                                                ; preds = %19, %2
  %8 = load i32, ptr %6, align 4
  %9 = load i32, ptr %4, align 4
  %10 = icmp slt i32 %8, %9
  br i1 %10, label %11, label %22

11:                                               ; preds = %7
  %12 = load ptr, ptr %3, align 8
  %13 = load i32, ptr %6, align 4
  %14 = sext i32 %13 to i64
```

```
int sum(int *vec, int length) {
    int total = 0;
    for (int i = 0; i < length; i++) {
        total += vec[i];
    }
    return total;
}
```

No pragma

No pragma

No pragma

Element type : int

Optimization level : -O0

- [ ] -msse3
- [ ] -mavx2
- [ ] -mavx512f
- [ ] -ffast-math
- [ ] -march=native

⚠ PlutoRunner.embed_display(x) is deprecated and will be removed in a future release. Use AbstractPlutoDingetjes.Display.@embed(x) instead.

LLVM Loop Vectorizer for a C++ vector

```
; ModuleID = '/tmp/jl_UfRUC8/main.c'
source_filename = "/tmp/jl_UfRUC8/main.c"
target datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-i128:128-f80:128-n8:16:32:64-S128"
target triple = "x86_64-unknown-linux-gnu"

; Function Attrs: noinline nounwind optnone uwtable
define dso_local i32 @sum(ptr noundef %0, i32 noundef %1) #0 {
  %3 = alloca ptr, align 8
  %4 = alloca i32, align 4
  %5 = alloca i32, align 4
  %6 = alloca i32, align 4
  store ptr %0, ptr %3, align 8
  store i32 %1, ptr %4, align 4
  store i32 0, ptr %5, align 4
  store i32 0, ptr %6, align 4
  br label %7

7:                                                ; preds = %19, %2
  %8 = load i32, ptr %6, align 4
  %9 = load i32, ptr %4, align 4
  %10 = icmp slt i32 %8, %9
  br i1 %10, label %11, label %22

11:                                               ; preds = %7
  %12 = load ptr, ptr %3, align 8
  %13 = load i32, ptr %6, align 4
  %14 = sext i32 %13 to i64
  %15 = getelementptr inbounds i32, ptr %12, i64 %14
```

```
32775.977f0
```

```julia
@btime cpp_sum($vec_float)
```

```
279.494 μs (0 allocations: 0 bytes)
```

```julia
cpp_sum(x::Vector{Cfloat}) = ccall(("c_sum", cpp_sum_float_lib), Cfloat,
(Ptr{Cfloat}, Cint), x, length(x));
```

```cpp
#include <vector>

int my_sum(std::vector<int> vec) {
  int total = 0;
  for (int i = 0; i < vec.size(); i++) {
    total += vec[i];
  }
  return total;
}

extern "C" {
int c_sum(int *array, int length) {
  std::vector<int> v;
  v.assign(array, array + length);
  return my_sum(v);
}}
```

No pragma

No pragma

No pragma

Element type : int

Optimization level : -O0

- [ ] -msse3
- [ ] -mavx2
- [ ] -mavx512f
- [ ] -ffast-math
- [ ] -march=native

```
⚠ PlutoRunner.embed_display(x) is deprecated and will be removed in a future rele
ase. Use AbstractPlutoDingetjes.Display.@embed(x) instead.
```

Tip

Easily call C++ code from Julia or Python by adding a C interface like the `c_sum` in this example.

LLVM Superword-Level Parallelism (SLP) Vectorizer

```
f (generic function with 2 methods)
```

```julia
f(a, b) = (a[1] + b[1], a[2] + b[2], a[3] + b[3], a[4] + b[4])
```

```julia
@code_llvm debuginfo=:none f((1, 2, 3, 4), (5, 6, 7, 8))
```

```
; Function Signature: f(NTuple{4, Int64}, NTuple{4, Int64})
define void @julia_f_34911(ptr noalias nocapture noundef nonnull sret([4 x i6
4]) align 8 dereferenceable(32) %sret_return, ptr nocapture noundef nonnull re
adonly align 8 dereferenceable(32) %"a::Tuple", ptr nocapture noundef nonnull
readonly align 8 dereferenceable(32) %"b::Tuple") #0 {
top:
  %0 = load <4 x i64>, ptr %"a::Tuple", align 8
  %1 = load <4 x i64>, ptr %"b::Tuple", align 8
  %2 = add <4 x i64> %1, %0
  store <4 x i64> %2, ptr %sret_return, align 8
  ret void
}
```

Inspection with godbolt Compiler Explorer

Example source

Interleaving

```
; ModuleID = '/tmp/jl_6OdPrE/main.c'
source_filename = "/tmp/jl_6OdPrE/main.c"
target datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-i128:128-f8
0:128-n8:16:32:64-S128"
target triple = "x86_64-unknown-linux-gnu"

; Function Attrs: noinline nounwind optnone uwtable
define dso_local void @add(ptr noundef %0, i32 noundef %1) #0 {
  %3 = alloca ptr, align 8
  %4 = alloca i32, align 4
  %5 = alloca i32, align 4
  %6 = alloca i32, align 4
  store ptr %0, ptr %3, align 8
  store i32 %1, ptr %4, align 4
  store i32 0, ptr %5, align 4
  store i32 0, ptr %6, align 4
  br label %7

7:                                                ; preds = %22, %2
  %8 = load i32, ptr %6, align 4
  %9 = load i32, ptr %4, align 4
  %10 = icmp slt i32 %8, %9
  br i1 %10, label %11, label %25

11:                                               ; preds = %7
  %12 = load ptr, ptr %3, align 8
  %13 = load i32, ptr %6, align 4
  %14 = sext i32 %13 to i64
  %15 = getelementptr inbounds i32, ptr %12, i64 %14
```

With `interleave_count(2)` and

- `-O1`, we see `!{!"llvm.loop.interleave.count", i32 2}` at the end but it hasn't ben applied
- `-O2`, we see the interleaving is applied

```
void add(int *a, int length) {
    int total = 0;
    for (int i = 0; i < length; i++) {
        a[i] = a[i] + 1;
    }
    return;
}
```

No pragma

No pragma

No pragma

Element type : int

Optimization level : -O0

- [ ] -msse3
- [ ] -mavx2
- [ ] -mavx512f
- [ ] -ffast-math
- [ ] -march=native

⚠ PlutoRunner.embed_display(x) is deprecated and will be removed in a future release. Use AbstractPlutoDingetjes.Display.@embed(x) instead.

Modern CPU architectures

Reorder buffer:

```
a = b + c;
d = a + c; // Needs to wait for a
e = b - c; // Independent from a
           // Can be executed before
```

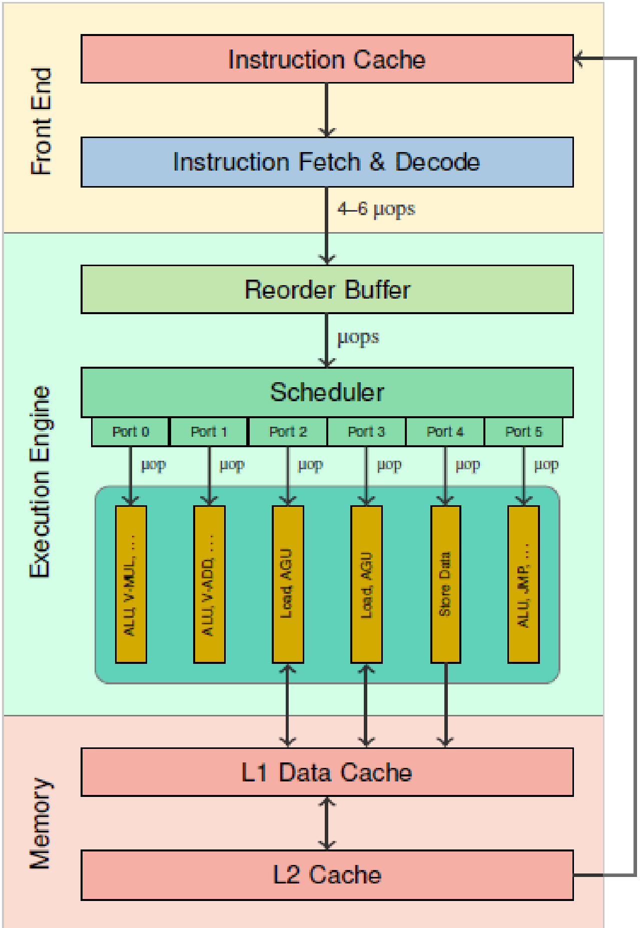
Execution ports

- Arithmetic Logic Unit (ALU)

```
a + b
```

- Address Generation Unit (AGU)
  - Load AGU in port 2 & 3
  - Store AGU in port 4

```
a[i] -> a + i * sizeof(...)
```



Why reordering ?

```
load  r1 ← a
load  r2 ← b
load  r3 ← c
load  r4 ← d
add   r5 ← r1 + r2
add   r6 ← r3 + r4
add   r7 ← r6 + 1
```

Assume

- L1 latency is 4 cycles
- L2 latency is 12 cycles
- `a` is missing from L1 cache but is in L2 cache

Without reordering

Cycle	ALU	ALU	Load	Load

With reordering

Cycle	ALU	ALU	Load	Load

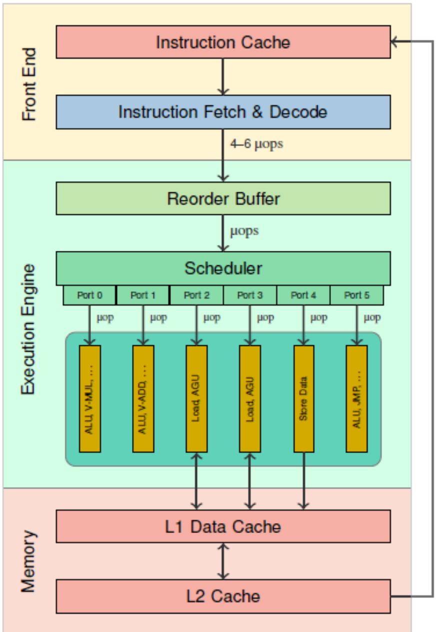
1			a	b
2			c	d
3-5			L2 ⌛	L1 ⌛
6-12			L2 ⌛	
13	r1 + r2	r3 + r4		
14	r6 + 1			

1			a	b
2			c	d
3-5			L2 ⌛	L1 ⌛
6		r3 + r4	L2 ⌛	
7		r6 + 1	L2 ⌛	
8-12			L2 ⌛	
13	r1 + r2			

Without unrolling

```
loop:
  load  r1 ← a[i]
  load  r2 ← b[i]
  add   r3 ← r1 + r2
  store c[i] ← r3
  add   i ← i + 1
  cmp   i, N
  jl   #loop ; Jump if Less
```

Cycle	ALU	ALU	Load	Load	Store	Jump
1			a[i]	b[i]		
2-4			L1 ⌛	L1 ⌛		
5	r1 + r2					
6	i + 1				c[i]	
7	i <? N					
8						#loop



Interleaving allowed by unrolling

```
loop2:
  load  r1 ← a[i]
  load  r2 ← b[i]
  load  r4 ← a[i+1]
  load  r5 ← b[i+1]
  add   r3 ← r1 + r2
  add   r6 ← r4 + r5
  store c[i] ← r3
  store c[i+1] ← r6
  add   i ← i + 2
  cmp   i, N-1
  jl  #loop2
  cmp   i, N
  jl  #loop ; tail case, no unrolling
```

Cycle	ALU	ALU	Load	Load	Store	Jump
1			`a[i]`	`b[i]`		
2			`a[i+1]`	`b[i+1]`		
3-4			`L1 ⌛`	`L1 ⌛`		
5	`r1 + r2`					
6		`r4 + r5`			`c[i]`	
7	`i + 2`				`c[i+1]`	
8	`i <? N-1`					
9						`#loop2`

Further notes

- Can be controlled with `#pragma clang loop interleave_count(4)` but the compiler usually already does a good job choosing the `interleave_count`
- This count depends on whether it is *compute bound* or *memory bound*, see next lecture where we discuss these.
- The above examples assumes the value of `a[i]` etc... are in L1 cache. If not (aka "L1 cache miss"), the latency for loading `a[i]`, will be much longer.

Additional practical limits include:

Limit	Typical value	Applies when

Load ports	2/cycle	Always — hard throughput ceiling
Load buffer (MOB)	~70-100 entries	All loads, hit or miss
Line fill buffers	~12 entries	L1 cache misses only
Reorder buffer	~200 μops	Window used for reordering

Fused instructions

Fused Multiply-Add (FMA)

```
; ModuleID = '/tmp/jl_YurY4M/main.c'
source_filename = "/tmp/jl_YurY4M/main.c"
target datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-i128:128-f8
0:128-n8:16:32:64-S128"
target triple = "x86_64-unknown-linux-gnu"

; Function Attrs: noinline nounwind optnone uwtable
define dso_local void @elementwise_muladd(ptr noundef %0, ptr noundef %1, ptr
noundef %2, i32 noundef %3) #0 {
  %5 = alloca ptr, align 8
  %6 = alloca ptr, align 8
  %7 = alloca ptr, align 8
  %8 = alloca i32, align 4
  %9 = alloca i32, align 4
  %10 = alloca i32, align 4
  store ptr %0, ptr %5, align 8
  store ptr %1, ptr %6, align 8
  store ptr %2, ptr %7, align 8
  store i32 %3, ptr %8, align 4
  store i32 0, ptr %9, align 4
  store i32 0, ptr %10, align 4
  br label %11

11:                                               ; preds = %33, %4
  %12 = load i32, ptr %10, align 4
  %13 = load i32, ptr %8, align 4
  %14 = icmp slt i32 %12, %13
  br i1 %14, label %15, label %36
```

```
void elementwise_muladd(int *c, int *a, int *b, int length) {
    int total = 0;
    for (int i = 0; i < length; i++) {
        c[i] += a[i] * b[i];
    }
    return;
}
```

No pragma

No pragma

No pragma

Element type : int

Optimization level : -O0

- [ ] -msse3
- [ ] -mavx2
- [ ] -mavx512f
- [ ] -ffast-math
- [ ] -march=native

```
⚠ PlutoRunner.embed_display(x) is deprecated and will be removed in a future rele
ase. Use AbstractPlutoDingetjes.Display.@embed(x) instead.
```

Explicit FMA: [ ]

Further readings

Slides inspired from:

- SIMD in Julia
- Demystifying Auto-vectorization in Julia
- Auto-Vectorization in LLVM
- Information on processor architecture
- Advanced books on optimization

The End

```
using SimpleClang, PlutoUI, PlutoUI.ExperimentalLayout, HypertextLiteral; Luxor,
StaticArrays, BenchmarkTools, PlutoTeachingTools
```

img (generic function with 3 methods)

```julia
begin
struct Path
    path::String
end

function imgpath(path::Path)
    file = path.path
    if !('.' in file)
        file = file * ".png"
    end
    return joinpath(joinpath(@__DIR__, "images", file))
end

function img(path::Path, args...; kws...)
    return PlutoUI.LocalResource(imgpath(path), args...)
end

struct URL
    url::String
end

function save_image(url::URL, html_attributes...; name = split(url.url, '/')[end],
kws...)
    path = joinpath("cache", name)
    return PlutoTeachingTools.RobustLocalResource(url.url, path,
html_attributes...), path
end

function img(url::URL, args...; kws...)
    r, _ = save_image(url, args...; kws...)
    return @htl("<a href=$(url.url)>$r</a>")
end

function img(file::String, args...; kws...)
    if startswith(file, "http")
        img(URL(file), args...; kws...)
    else
        img(Path(file), args...; kws...)
    end
end
end
```

```
qa (generic function with 2 methods)
```

```
begin
function qa(question, answer)
    return @htl("<details><summary>$question</summary>$answer</details>")
end
function _inline_html(m::Markdown.Paragraph)
    return sprint(Markdown.htmlinline, m.content)
end
function qa(question::Markdown.MD, answer)
    # `html(question)` will create `<p>` if `question.content[]` is `Markdown.Paragraph`
    # This will print the question on a new line and we don't want that:
    h = HTML(_inline_html(question.content[]))
    return qa(h, answer)
end
end
```

```
function Luxor.placeimage(url::URL, pos; scale = 1.0, centered = true, kws...)
    r, path = save_image(url; kws...)
    if r.mime isa MIME"image/svg+xml"
        img = Luxor.readsvg(path)
    else
        img = Luxor.readpng(path)
    end
    Luxor.gsave()
    Luxor.scale(scale)
    Luxor.placeimage(img, pos / scale; centered)
    Luxor.grestore() # undo `Luxor.scale`
    return
end
```

```
CenteredBoundedBox (generic function with 1 method)
```

```
function CenteredBoundedBox(str)
    xbearing, ybearing, width, height, xadvance, yadvance =
        Luxor.textextents(str)
    lcorner = Luxor.Point(xbearing - width/2, ybearing + height/2)
    ocorner = Luxor.Point(lcorner.x + width, lcorner.y + height)
    return Luxor.BoundingBox(lcorner, ocorner)
end
```

```
boxed (generic function with 1 method)
```

```julia
function boxed(str::AbstractString, p; hue = "lightgrey")
    Luxor.gsave()
    Luxor.translate(p)
    Luxor.sethue(hue)
    Luxor.poly(CenteredBoundedBox(str) + 5, action = :stroke, close=true)
    Luxor.sethue("black")
    Luxor.text(str, Luxor.Point(0, 0); halign = :center, valign = :middle)
    #settext("<span font='26'>$str</span>", halign="center", markup=true)
    Luxor.origin()
    Luxor.grestore() # strokecolor
end
```